Adaptive Composite Feature Generation for Object Detection in Remote Sensing Images

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***Abstract*— Object detection in remote sensing images identifies and extracts the acquired Earth surface information, providing data support and research basis for multiple fields. Remote sensing image object detection based on knowledge distillation (KD) can transfer the knowledge of a large teacher model to a smaller student model, achieving the effect of low parameter volume and high accuracy. Mainstream methods directly imitate teacher features to improve student performance, ignoring the generation of high-ranking features through teacher features instructing student feature maps in this knowledge transfer process. In this article, an adaptive composite feature generation (ACFG) strategy is proposed to achieve end-to-end trainable KD for object detection in remote sensing images, in which the robustness of feature points under composite masks is improved through adaptive feature mapping. In particular, a composite mask generator (CMG) module is proposed to select student instance-related features and point background features. Furthermore, a global and local projection layer (GLPL) module is proposed to connect the local information and global information of the feature map under the mask generator to adaptively realize the global recovery mapping of the feature map with partial feature points. Finally, balanced decoupling loss (BDL) is improved to handle foreground and background loss separately, so that the two decoupled features can better enable the student model to learn instance-related information. Note that the proposed ACFG is capable of conducting KD for both single-stage and two-stage object detectors. Experimental results using both anchor-based and anchor-free detectors on the DIOR dataset and DOTA dataset demonstrate that the proposed ACFG clearly achieved better performance than several state-of-the-art (SOTA) algorithms for KD.**

***Index Terms*— Knowledge distillation (KD), object detection, remote sensing image, self-attention.**

I. Introduction

With the continuous development and progress of remote sensing technology, the resolution and coverage area of remote sensing image data [1], [2] collected by remote sensing platforms such as satellites and aviation have continued to increase, making remote sensing data an important tool for obtaining information on the Earth's surface [3], [4], [5]. Object detection is one of the important links in remote sensing data processing, which provides an important data support and application basis for the fields of geographic information systems, urban planning, environmental monitoring, and so on.

Various object detectors [6], [7], [8], [9] have been proposed, which achieve high accuracy on large datasets and show good experimental results with the development of deep convolutional neural networks. It is also widely used in practical engineering, such as autonomous driving, personnel search and rescue, and various visual tasks based on UAV platforms. The existing object detectors [10], [11] are mainly based on convolutional neural networks. The Facebook research team proposed the first transformer-based object detection model DETR [12], and in recent years transformer-based object detectors have also been gradually developed. However, these models often require high computing power resources and large storage space, are less practical, and encounter difficulties in platform deployment. Therefore, it is of great importance to study lightweight real-time object detectors with high performance and fast operation speed. In order to implement real-time lightweight models, there are several mainstream algorithms in the current stage, including quantization [13], lightweight model design [14], pruning [15], and knowledge distillation (KD) [16]. These algorithms provide solutions for real-time object detection, increasing the practical value of object detectors.

In this article, we focus on KD [17] due to its powerful knowledge transfer and accelerated training capabilities, which is a technique for model compression and transfer learning designed to transfer knowledge from a complex teacher model to a simple student model. It is well-known that KD algorithms were initially introduced to solve image classification tasks [17], which limited their applicability to various object detectors, especially for remote sensing images.

Compared with ordinary visible images, remote sensing images have complex background information, and objects often exhibit multiscale characteristics, which pose a challenge for localization. Different positions in the image, such as background, foreground, center, or boundary, may contribute differently to the detection performance, which makes the KD task more challenging. To address these issues, some improvements have been made to existing KD algorithms for object detection. Wang et al. [16] employed fine-grained mask filtering to background regions, allowing distillation only from the relevant foreground regions. Guo et al. [18] proposed to decouple the foreground and background and

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introduced a balance factor to calculate the distillation loss. Zhixing et al. [19] proposed a novel feature-richness score (FRS) method based on the aggregated classification score to select appropriate regions for distillation. Yang et al. [20] proposed a method to distill the foreground and background, respectively, and introduced focal distillation and global distillation. Soon after, they pointed out that the teacher model can improve the student model's feature representation by guiding the student model's recovery, which discards the original concept of having the features of the student model mimic the teacher model [21].

Current distillation algorithms are dedicated to selecting suitable distillation regions by various methods, but they have gradually proved to be unfavorable for improving the detection performance as they only focus on extracting the foreground region while ignoring the background information. Many existing algorithms mainly focus on the foreground, either treating the foreground and background uniformly or decoupling the background independently, and do not make full use of the correlation between the foreground and background information to improve the robustness of the feature points. In order to make distillation features take foreground and background features into account, a composite mask generator (CMG) module that introduces target sizes and background information is proposed to improve the diversity of distillation features and the robustness of each feature point.

Due to the different complexity of the feature extraction network of the teacher model and the student model, the size and number of channels of the feature maps in the feature fusion network are also different. Therefore, exploring the mapping relationship between the student model feature maps and the teacher model feature maps is also an important part of feature distillation, which plays an important role in how to select appropriate student features for distillation and the calculation of distillation loss. The existing methods [18], [20], [21] all adopt the model structure based on CNNs to realize the mapping between the two feature maps, which only focus on the local information of the feature maps, and some manually-designed multibranch distillation methods are affected by human-defined hyperparameters. As a result, the student model cannot fully grasp the contextual features that describe different objects and backgrounds. On the one hand, the transformer architecture utilizes a multilayer perceptron (MLP) to model distant correlations between tokens in a sequence, which enables KD to capture global information from images [22], [23]. On the other hand, deformable convolution (DCN) [24] employs different convolution kernel shapes for different targets, which can enhance the local features of the target. Therefore, the global and local projection layer (GLPL) is proposed to implement feature mapping, which is used as a special hyperparameter model to guide the student model to achieve better detection results. GLPL introduces an attention mechanism for constructing global relationships and DCN for local information extraction to realize adaptive mapping of feature maps. In addition, adaptive composite feature generation (ACFG) proposes an improved loss called the balanced decoupling loss (BDL), which introduces instance sizes in the foreground matrix to emphasize different target sizes during knowledge transfer. As a result, BDL can better measure the feature differences as well as achieve feature transfer with small targets.

The main contributions are summarized as follows.

1) An end-to-end adaptive composite distillation trainable framework is proposed to make full use of global and local information to achieve knowledge transfer and improve the performance of student models. On both DIOR and DOTA datasets, the student models trained with ACFG demonstrate outstanding results while maintaining low complexity.
2) The CMG is proposed to generate diversified composite masks to realize knowledge transfer between foreground and background information during the training process, by which target feature information and contextual information can be utilized to enhance the robustness of student features.
3) In order to construct the mapping relationship between the student model and the teacher model, GLPL is proposed to construct the global information correlation and enhance the local information, which adaptively implements the global and local mapping of the feature maps during the knowledge refinement process.
4) Considering the differences between foreground and background, BDL is proposed to introduce instance sizes in the foreground mask to emphasize the features of small targets and decouple the background separately to more accurately measure the differences between teacher and student models.

The remaining sections of this article are organized as follows. First, Section II briefly reviews some recent works related to object detection and KD. Second, Section III describes the details of our proposed method. Third, in Section IV, a series of experiments and results Are illustrated and analyzed to compare our model with several state-of-the-art (SOTA) methods. The use of ground truth, the improvement of distillation losses, and the applicability of different remote sensing data are also discussed. Finally, the conclusion is drawn in Section V.

II. Related Work

A. Object Detection

Object detection is one of the important research directions in computer vision. The current object detection algorithms mainly include two-stage detectors [6], [25] and single-stage detectors [7], [8], [9], which can also be divided into anchor-based detectors [7], [9], and anchor-free detectors [8]. The two-stage object detection algorithm transforms the detection problem into a classification problem of the local image in the generated proposed region. In the first stage, this kind of algorithm generates region proposals, classifies the foreground and background in the image, and deduces the result of the foreground region in the image. In the second stage, classification and regression are performed on the content in the region of interest corresponding to the feature map in the convolutional neural network, which is usually a multiclassification task. The one-stage object detection algorithm treats the object detection task as a regression task on the whole image

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and directly predicts the category and bounding box according to the output. The above object detection algorithms are mainly based on deep convolutional neural networks. In addition, the transformer has been widely used in the field of object detection due to its good performance.

Many object detection algorithms rely on strong backbones such as VGGs [26], ResNets [27], and DarkNets [9] to extract features from images [28]. As a result, recent research has focused on creating lightweight backbones [14], [29] to address the need for faster and more efficient object detection models. However, lightweight detectors often do not have good performance for remote sensing images with complex backgrounds and multiscale objects.

The complex background and noise of remote sensing images are the major challenges of object detection. The instances in remote sensing images have the characteristics of large aspect ratio, inconsistent direction, multiscale, and large intraclass scatter. Focusing on the characteristics of remote sensing images, many object detection algorithms [30] have been proposed and achieved better detection performance. Xie et al. [31] constructed a lightweight RPN network to extract region proposals, and detected target categories and locations by rotated RoI Alignment module. Yang et al. [32] employed feature point interpolation to construct a feature refinement module for feature alignment, and designed derivable approximate SkewIoU as a loss function. Han et al. [33] designed AlignConv to encode orientation information, and employed an anchor refinement module to generate high-quality anchors and align convolutional features. Han et al. [34] introduced rotational isovariance and rotational invariance to target detection networks, and the alignment module extracted features adaptively according to the orientation of the RoI. Wang et al. [35] proposed a novel multiscale feature enhancement network (MFENet) to improve detection performance by constructing hybrid feature enhancement and semantic feature guidance modules. Dong et al. [36] proposed a new multiscale deformable attention module (MSDAM) to generate attention maps from feature maps with multiscale deformable perceptive domains, which can better fit remote sensing objects of various shapes and sizes.

B. Knowledge Distillation

Hinton et al. [17] proposed the concept of KD, which was initially applied to extract knowledge from a large, complex model by teaching a smaller, simpler model, and achieved good results in classification tasks. Through KD, the student model can learn the prediction ability and generalization ability of the teacher model, so as to achieve similar performance as the teacher model while maintaining a low complexity of the student model. KD can be divided into logit-based [17], feature-based [37], [38], and relation-based [39], [40] methods. KD was first proposed to be applied to classification tasks. Compared with the classification task, the object detection task needs to predict the object's position as well as recognize its category, which generally adopts more complex network structures and optimization objectives. Therefore, it is more challenging to establish an effective KD framework for object detection tasks.

As we all know, the feature maps of the object detector contain rich feature information, which can effectively improve the effect of KD. FitNet [37] was first proposed in the image classification task to achieve KD by extracting feature maps. When it is applied to the object detector, it fails to transfer the location and size of targets from the teacher model to the student model. Many subsequent KD algorithms for object detection are dedicated to introducing positioning information and category information into the distillation process.

For the first time, Chen et al. [41] successfully implemented KD for multiclass object detection problems, and they introduced the output of intermediate layers into the final classification and regression results. Yang et al. [20] proposed that the feature points and channels noticed by teachers and students are completely different, so they explored the key areas of distillation and proposed focal and global distillation (FGD) accordingly, introducing attention mechanisms to achieve local distillation. Since manual selection cannot fully learn useful information, many attention-based methods [42], [43] have been proposed to select local features and channel features on the feature map. There are also some algorithms [44], [45] that introduce generative adversarial networks (GANs) into feature distillation to force the student model to generate feature maps that are consistent with the teacher model. In addition, since the remote sensing object is connected to the background, various methods [18], [19], [46] have been proposed for separately decoupling foreground and background. Yang et al. [21] clarified that the teacher model can improve the performance of the student model by guiding the feature recovery of the student model, and proposed a masked generative distillation (MGD) method. Li et al. [47] combined relationship-based and feature-based knowledge distillation and created imitation masks based on candidate RoIs generated by a region proposal network. It is important to pay attention to the area and background around the object in the distillation process. Current imitation masks usually ignore information from the background because the network mainly focuses on objects or specific misleading regions. To further improve detection performance, we create local target masks and global random masks which facilitate shifting attention to other important patches that contain local information.

III. Proposed Method

A. Overall Framework

The framework of our proposed ACFG is illustrated in Fig. 1, which consists of a CMG module, a GLPL module, and BDL as KD loss. First, in order to implement feature selection from the student model, CMG uses the foreground masks to refine rich target information and the random masks to provide background information, which acts as a kind of contextual information to provide additional hints for detecting the target. Second, in order to enhance the local and global expression of student model features, the GLPL combines the self-attention mechanism and DCN for feature mapping, which allows the network to better learn the global correlation as well as the target shape features. Moreover, GLPL acts as an intermediate

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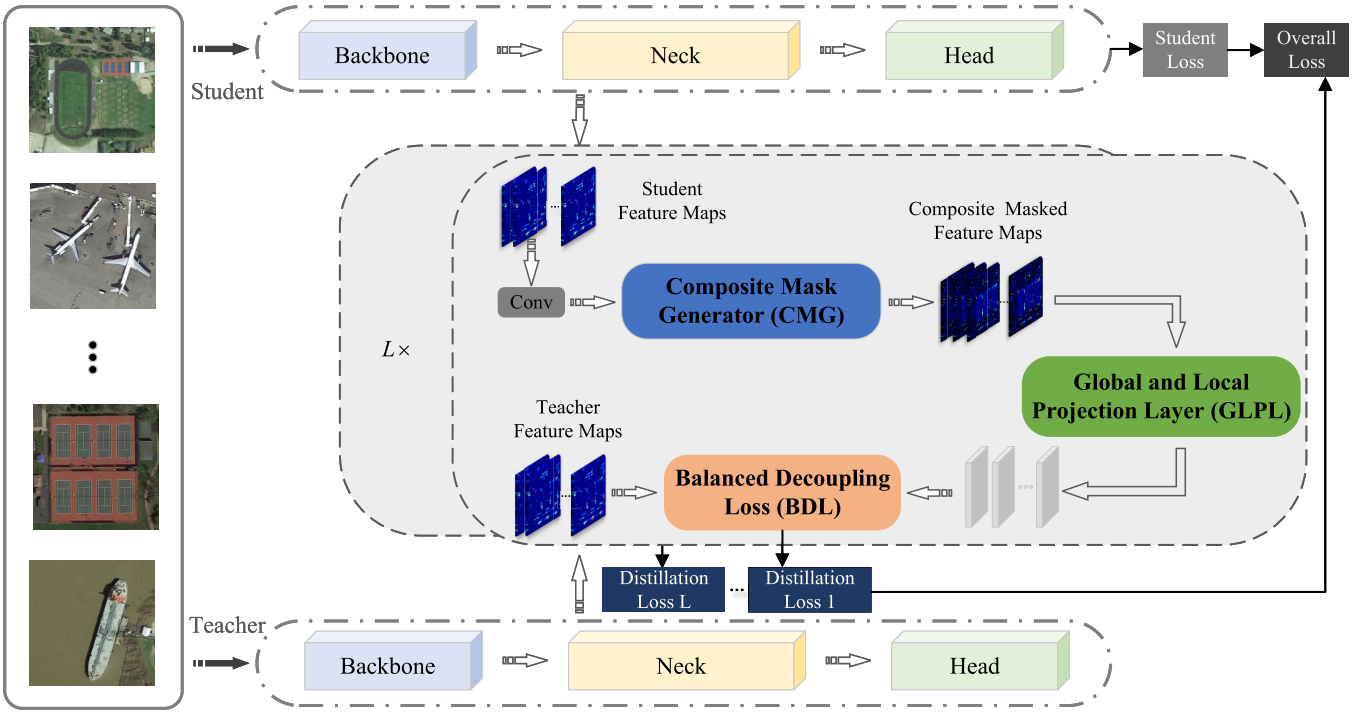


Fig. 1. Overall structure of our proposed ACFG. The input images first go to the teacher and student backbone networks for feature extraction, respectively, then to the feature fusion network, and finally to the detection head. We use CMG to select the feature for distillation, employ the GLPL module to adaptively learn the global and local features of the teacher, and compute the loss of student and teacher features.

layer capable of mitigating the overfitting between the two feature maps. Third, taking into account the differences between foreground and background information and the multiscale characteristics of instances in remote sensing images, BDL is proposed to measure the loss of two feature maps accurately. According to the probability mask matrix generated by the instance size of each pixel in the image, the foreground and background losses are calculated, respectively, to obtain the final KD loss.

B. CMG Module

As we all know, most of the mainstream object detection models are based on CNNs, and different features have different receptive fields and information representation capabilities. Existing methods claimed that partial feature points can recover the complete feature map since the mapped points on the feature map already contain the information of neighboring points to some extent [21]. However, remote sensing images contain multiple multiscale instances, and all of these instance features contribute to the object detectors to distinguish the foreground from the background to achieve localization. Therefore, we provide a composite structure that superimposes the foreground and background to obtain a composite mask. The proposed CMG module employs an instance branch to emphasize the instance sizes and introduces a random mask branch as a supplement to the context information. Then, instead of processing local and global information separately, a global–local feature distillation paradigm is performed through the same projection layer, using different background information as well as all foreground information in each round, so that the network can better learn the global and local features related to the target. It is worth noting that the mask generator only requires ground truth during the training phase as a medium to enhance student network learning. In the validation and testing phases, the ground truth is not required to be provided to the mask generator and only the regular student network model is used for detection.

We transfer the features generated by each layer of the neck to achieve feature-based KD. The structure of CMG is shown in Fig. 2. The mask calculation process is summarized as follows.

First, assume that the feature maps of students and teachers are represented as $F^s \in \mathbb{R}^{N\times C_s\times H_s\times W_s}$ and $F^t \in \mathbb{R}^{N\times C_t\times H_t\times W_t}$, respectively, where N represents the batch size, H_s and W_s represent the height and width of the student feature map, and H_t and W_t represent the height and width of the teacher feature map. In addition, C_t and C_s denote the number of channels of the teacher feature map and student feature map. Since the channels are not unified, that is, C_s is not equal to C_t, 1×1 Conv is used to change the student features to obtain features of the same dimension. If the height and width of the student feature map and the teacher feature map are inconsistent, use the resize operation to make the height and width of the student feature map consistent with the teacher feature map. The processed features are represented as $F^{\text{st}} \in \mathbb{R}^{N\times C_t\times H_t\times W_t}$. Due to the recovery properties of some feature points and ensuring the randomness of the selected points, we set a random mask branch to cover the student's features to provide background information to assist in localization. The same random mask is used for all channels of the feature map of each image to achieve consistency in constructing global and local information across channels. Thus, the first channel

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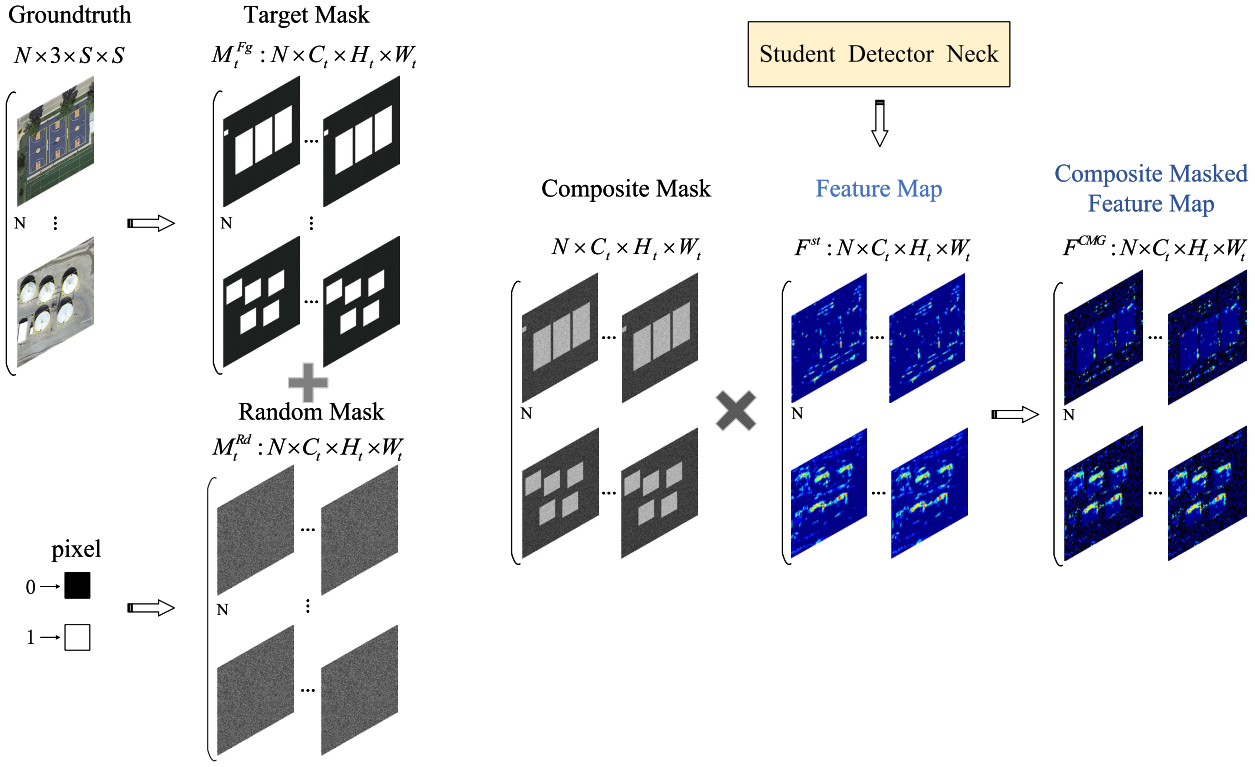


Fig. 2. Structure of proposed CMG module.

is taken to illustrate the mask generation

$$M_{n,1,i,j}^{\mathrm{rd}} = \begin{cases} 0, & a_{n,1,i,j} < \lambda \\ m, & \text{otherwise} \end{cases} \tag{1}$$

where $a_{n,1,i,j}$ is a random number in $[0, m)$ that conforms to the uniform distribution, and n, i, and j are the number of images in a batch and the horizontal and vertical coordinates of the feature map, respectively. In addition, λ represents a hyperparameter that sets the threshold. m represents the upper bound of uniform distribution, which is set to 1. $M_{n,1,i,j}^{\mathrm{rd}}$ is copied C_t times in the channel direction to generate the final random mask $M_t^{\mathrm{Rd}} \in \mathbb{R}^{N\times C_t\times H_t\times W_t}$.

Second, considering the imbalance of foreground and background information in object detection tasks, we set a target mask branch to separate the background and foreground. Background feature points can be regarded as independent random variable points, but for different foreground instance sizes, we set the mask of the object region according to the object size to enhance the extraction of the target, which can be formulated as follows:

$$M_{n,1,i,j}^{\mathrm{fg}} = \begin{cases} \dfrac{1}{H_{\mathrm{fg}} \times W_{\mathrm{fg}}}, & \text{if } b(n,1,i,j) \in \mathrm{fg} \\ 0, & \text{otherwise} \end{cases} \tag{2}$$

where H_{fg} and W_{fg} denote the height and width of the smallest ground truth box on each feature point, and $b(n, 1, i, j)$ represents the feature point on the feature map. In addition, fg represents the corresponding ground truth box area on the feature map. All channels of the feature map of each image use the same instance mask in order to select all object features. Therefore, $M_{n,1,i,j}^{\mathrm{fg}}$ is copied C_t times in the channel dimension to generate the final instance mask $M_t^{\mathrm{Fg}} \in \mathbb{R}^{N\times C_t\times H_t\times W_t}$.

Then, we use the composite mask to cover the student feature map and set two hyperparameters to balance the two branches. This part operates as follows:

$$x_l = F^{\mathrm{CMG}} = \left(\beta \times M_t^{\mathrm{Rd}} + (1-\beta) \times M_t^{\mathrm{Fg}}\right) \cdot F^{\mathrm{st}} \tag{3}$$

where β is the parameter that balances the two branches, which is set to the expectation of a uniform distribution, and L is the number of groups of feature maps, l is the serial number of the group of feature maps. For the neck of the object detector, L usually takes 3–5.

C. GLPL Module

Current distillation algorithms usually improve students' performance by imitating the teacher's output or feature maps, ignoring the characteristics of using partial feature points and simple networks to generate complete feature maps. After passing through the feature map of the convolutional neural network, each point on the feature map already contains the surrounding local information, so some points on the feature map have the ability to reconstruct surrounding feature points. Previous studies have shown that adding a projection layer after the output features of the student model is beneficial to alleviate the overfitting between two feature maps. Such simple blocks used by existing algorithms are usually multiple 3×3 Conv layers and adding an activation layer relu between every two convolution layers, as well as adding some residual structures and various activation functions. Even if these modules handle the feature map globally, they only focus on

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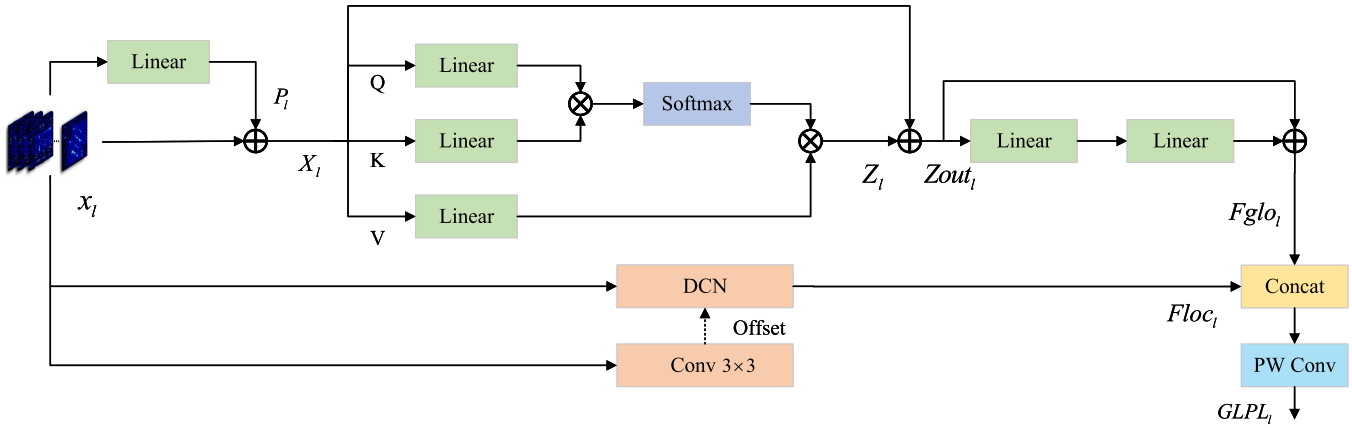


Fig. 3. Architecture of the proposed GLPL. We use the self-attention mechanism to construct a global relational branch and use DCN to enhance local features and construct a local feature branch.

the local information of the image through the sliding of the convolution block, ignoring the global relationships.

Nonlocal operations [48] have been proven to be of great importance for remote sensing object detection tasks by modeling long-term dependence on different spatial locations. Therefore, we propose the GLPL module to implement feature mapping so that the student feature map fully learns the information from the teacher feature map. Specifically, we first use the self-attention mechanism to realize global relationship modeling and improve the global information extraction ability of the model. Since DCN owns better feature extraction capability for irregular targets because of the diversity of convolution kernel shapes, we adopt it for local information extraction. Then, we concatenate the extracted global and local features and finally use channel convolution to achieve global and local information reduction and interaction. In the process of multilevel feature distillation, we use a set of projection layers to implement the mapping of student feature maps and teacher feature maps.

As shown in Fig. 3, GLPL consists of a global feature branch and a local feature extraction branch. The mapped features build global relationships and improve local information representation capabilities. In this section, we explore the impact of different generating blocks on detection performance. The structure of the GLPL module is described in detail below.

In the global feature extraction, for input $x_l \in \mathbb{R}^{N\times C_l\times H_l\times W_l}$, it is first compressed into a 3-D tensor, combining W and H of the feature map into one dimension. In order to meet the input requirements of self-attention, the global pixel dimension is shifted to the first dimension, a dimension is added in front of the above 3-D tensor, and then the first and fourth dimensions are exchanged and the last dimension is deleted to achieve batch processing of image features. The position embedding $P_l \in \mathbb{R}^{D_l\times N\times C_l}$ is generated by x_l through the linear layer, and the final 3-D tensor $X_l \in \mathbb{R}^{D_l\times N\times C_l}$ is used as input to linear layers, which can be formulated as follows:

$$P_l = \text{Linear}(x_l) \tag{4}$$

$$X_l = x_l + P_l \tag{5}$$

$$X = \{X_1, X_2, \ldots, X_L\} \tag{6}$$

$$D_l = H_l \times W_l \tag{7}$$

where D_l is the length of the input sequence, and C_l is the embedding dimension, C_l also represents the channel number of feature maps from different layers of the teacher object detector's neck.

Each group goes through a transformer layer that contains an attention block and a feedforward block. The process can be formulated as follows. First, the input 3-D vector $X_l \in \mathbb{R}^{D_l\times N\times C_l}$ is mapped to three different matrices, namely, query (Q), key (K), and value (V) by linear transformations. The formula is as follows:

$$Q_l = X_l W_l^Q, \quad K_l = X_l W_l^K, \quad V_l = X_l W_l^V \tag{8}$$

where W_l^Q, W_l^K, and $W_k^V \in \mathbb{R}^{d\times d}$ are the learned parameter matrices. Second, in attention block, the process can be unified into a single formula as follows:

$$Z_l = \text{Attention}(Q_l, K_l, V_l) = \text{Softmax}\left(\frac{Q_l \times K_l}{\sqrt{d_l}}\right) \times V_l. \tag{9}$$

A residual connection is added to each attention block, which strengthens the flow of information in order to achieve higher performance. The formula is as follows:

$$\text{Zout}_l = Z_l + X_l \tag{10}$$

where $\text{Zout}_l \in \mathbb{R}^{D_l\times N\times C_l}$. Third, a feed-forward network (FFN) is applied after the attention layers. It consists of two linear transformation layers and a residual connection. This part can be denoted as the following function:

$$\text{FFN}_l = \text{FFN}(\text{Zout}_l) = W_2 W_1(\text{Zout}_l) \tag{11}$$

$$\text{Fout}_l = \text{FFN}_l + \text{Zout}_l \tag{12}$$

where W_1 and W_2 represent the two-parameter matrices of the two linear transformation layers. Finally, $\text{Fout}_l \in \mathbb{R}^{D_l\times N\times C_l}$ is reshaped into a 4-D vector $\text{Fg}_l \in \mathbb{R}^{N\times C_l\times H_l\times W_l}$. In the local feature extraction, for input $x_l \in \mathbb{R}^{N\times C_l\times H_l\times W_l}$, it is first fed to the convolutional layer to get the offset, and then the convolutional kernel with offset is used for feature extraction. Finally, we concatenate the obtained global branch features Fg_l and local branch features Fl_l, and use pointwise convolution for reducing channel dimensions to get GLPL_l

$$\text{Offset}_l = \text{Conv3} \times 3(x_l) \tag{13}$$

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$$\mathrm{Fl}_l = \mathrm{DCN}(x_l, \mathrm{Offset}_l) \tag{14}$$

$$\mathrm{GLPL}_l = \mathrm{PWConv}\big(\mathrm{Concat}\big(\mathrm{Fg}_l, \mathrm{Fl}_l\big)\big) \tag{15}$$

where DCN denotes deformable convolution for local feature extraction and PWconv denotes pointwise convolution for a reduction in the number of channels.

D. BDL and Overall Loss

The knowledge transferred from teacher feature maps can significantly improve the performance of student detectors. Therefore, it is crucial to measure the loss between teacher feature maps and mapped student feature maps. Previous distillation losses usually select foreground areas and ignore background information, or treat all areas equally, which ignores the differences between foreground and background. Shallow features are extracted from the first few layers of the network owning large feature maps and rich texture information, while deep features are close to the output of the feature extraction network, which owns smaller feature maps and rich semantic information. In the feature fusion part of the target detection network, a feature pyramid network (FPN) is constructed over both shallow and deep features, and the interaction of shallow and deep features is achieved by using up-sampling, down-convolution, and splicing operations. Since different levels of features own different receptive fields, different amounts of information, and different depths, in this article, different multilevel features of FPN are used for feature distillation to detect targets of different sizes using shallow and deep features.

The background region of the feature map can provide the feature of the scene where the target is located to achieve auxiliary detection, while large instances take up more loss due to more pixels, affecting the refinement of small objects. Therefore, BDL introduces a mask according to the size of instances to compute foreground loss. Particularly, a binary mask is used to decouple the background features individually to compute the background loss taking into account the consistency of the background. Since BDL is computed only on the feature map obtained from the neck of the detector, it can be easily applied to different detectors to balance the differences between the foreground and the background, which is described in detail below.

Given the intermediate features of size $H \times W$, we first generate an object mask M^{fg} and a background mask M^{bg} based on the ground truth box. The object mask is calculated according to (2). For different levels of features, we distribute each ground truth box to its corresponding level and generate the object mask and background mask for each level accordingly. Then, we can use the generated masks to decouple the neck features. The BDL for feature distillation is formulated as follows:

$$L_l^{\mathrm{fg}} = \sum_{n=1}^{N}\sum_{i=1}^{C}\sum_{j=1}^{H}\sum_{j=1}^{W}\big(F^t_{n,k,i,j} - f\big(F^{\mathrm{st}}_{n,k,i,j}\big)\big)^2 \cdot M^{\mathrm{fg}}_{n,k,i,j} \tag{16}$$

$$L_l^{\mathrm{bg}} = \sum_{n=1}^{N}\sum_{i=1}^{C}\sum_{i=1}^{H}\sum_{i=1}^{W}\big(F^t_{n,k,i,j} - f\big(F^{\mathrm{st}}_{n,k,i,j}\big)\big)^2 \cdot M^{\mathrm{bg}}_{n,k,i,j} \tag{17}$$

$$M^{\mathrm{bg}}_{n,1,i,j} = \begin{cases} 1, & \text{if } b(n,1,i,j) \in \mathrm{bg} \\ 0, & \text{otherwise} \end{cases} \tag{18}$$

$$L_l^{\mathrm{fea}} = W_{\mathrm{fg}} \times \sqrt{L_l^{\mathrm{fg}}} + W_{\mathrm{bg}} \times \sqrt{L_l^{\mathrm{bg}}} \tag{19}$$

where bg represents the background area on the feature map, W_{fg} and W_{bg} represent the hyperparameters that tune the foreground and background losses, f represents the general mapping function, and N, C, H, and W denote the batch size, the channel number, height, and width of the feature map, respectively. In our proposed method, f contains three parts: adaptation layers, CMG, and GLPL

$$f\big(\mathrm{Fea}^s_{n,k,i,j}\big) = \mathrm{GLPL}(F_l). \tag{20}$$

The overall loss is the sum of the student model's original loss and the distillation loss, and it is used for backpropagating through the student model to adjust its parameters

$$L_{\mathrm{all}}^{\mathrm{fea}} = \sum_{l=1}^{L} L_l^{\mathrm{fea}} \tag{21}$$

$$L_{\mathrm{all}} = \alpha \times \ L_{\mathrm{all}}^{\mathrm{fea}} + L_{\mathrm{org}} \tag{22}$$

where α is a hyperparameter to balance the two losses and L_{org} represents the original detection loss of the student model. The detection loss includes the classification loss and the regression loss

$$L_{\mathrm{org}} = \frac{1}{N_{\mathrm{cls}}}\sum_t\big(L_{\mathrm{cls}}\big(p_i, p_i^*\big)\big) + \frac{1}{N_{\mathrm{reg}}}\sum_t\big(L_{\mathrm{reg}}\big(t_i, t_i^*\big)\big) \tag{23}$$

where t represents the index of an anchor in a mini-batch, p_i denotes the predicted probability of anchor t being an object, and p_i^* is the ground truth label. t_i denotes a vector representing the four parameterized coordinates of the predicted bounding box, while t_i^* is the ground truth box associated with a positive anchor. L_{cls} is the classification loss and L_{reg} is the regression loss.

IV. Experimental Results and Analysis

A. Datasets

To evaluate the effectiveness and robustness of our proposed distillation method, we conduct experiments on two large-scale remote sensing object detection datasets, including DIOR [1] and DOTA [2]. The DIOR dataset includes 23 463 images, which consists of 20 types of objects with various dimensions and orientations. We use 5862 images for training, 5863 images for validation, and 11 738 images for testing. The DOTA dataset includes 2806 images of various sizes, which consists of 16 types of instances. For the DOTA dataset, we use the default partition for experiments. We split the original dataset with DOTA devikit and the images are cropped into the 1024×1024 pixels patches, where 100 pixels overlap with neighboring patches.

B. Experimental Settings

In our proposed ACFG experiments and comparative experiments with other KD methods, RetinaNet [7] (anchor-based one-stage detector), Faster RCNN [6] (anchor-based two-stage detector), and FCOS [8] (anchor-free one-stage detector)

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TABLE I
RESULTS OF THE STUDENT MODEL, TEACHER MODEL, AND ACFG BASED ON DIFFERENT DETECTORS ON THE DIOR AND DOTA

Method	DIOR				DOTA			
	mAP	FPS	Params (M)	FLOPs (G)	mAP	FPS	Params (M)	FLOPs (G)
RetinaNet (T)	60.9	11	94.22	178.37	50.3	10	94.14	177.66
RetinaNet (S)	56.6	17	36.50	85.05	47.9	18	36.42	84.34
ACFG	60.5	17	36.50	85.05	50.6	19	36.42	84.34
Faster RCNN (T)	60.5	14	60.21	121.53	52.6	15	60.19	121.51
Faster RCNN (S)	58.0	18	41.22	91.10	52.0	19	41.20	91.08
ACFG	60.4	18	41.22	91.10	53.1	19	41.20	91.08
FCOS (T)	63.1	11	89.65	174.22	53.6	11	89.64	174.14
FCOS (S)	58.2	15	32.88	59.29	33.5	15	32.87	59.22
ACFG	62.8	16	32.88	59.29	53.2	16	32.87	59.22
YOLOv3 (T)	61.6	25	61.63	77.70	56.4	27	61.60	77.67
YOLOv3 (S)	49.3	32	3.69	6.62	50.9	36	3.68	6.61
ACFG	56.1	32	3.69	6.62	52.0	36	3.68	6.61
YOLOX (T)	73.9	23	25.29	36.78	66.2	26	25.29	36.78
YOLOX (S)	70.8	29	8.95	13.34	64.0	32	8.94	13.34
ACFG	71.9	33	8.95	13.34	65.1	35	8.94	13.34

are used as the benchmark detection frameworks. We use ResNet101 and ResNet50 as the backbones for teacher detectors and student detectors, respectively. In addition, we conducted experiments on YOLOv3 [9] and YOLOX [49]. For YOLOv3, MobileNet and DarkNet are selected as the backbone network for students and teachers. For YOLOX, CSPDarknet is selected as the backbone network for students and teachers, and the deepen factor and widen factor are set to $\{0.33, 0.5\}$, $\{0.67, 0.75\}$, respectively.

The NVIDIA GeForce RTX 3080 GPU with 10 GB of memory and the NVIDIA GeForce RTX 3090 GPU with 20 GB of memory are used to accelerate the experiments. Our implementation is on the basis of mmdetection [50] and pytorch [51]. In order to get better performance and faster convergence, all the backbones are initialized with the pretrained weights on ImageNet [52]. We implement all the experiments with default configure files and the input image size is 640×640. The network is trained with 24 epochs using stochastic gradient descent (SGD). The initial learning rate is 0.003 for YOLOv3 and 0.01 for the other object detectors, and the momentum and weight decay are set to the default value of mmdetection. For fairness, the batch size for each type of object detector is set to be the same. On the DIOR dataset, the batch sizes for RetinaNet, Faster RCNN, FCOS, YOLOv3, and YOLOX are 4, 1, 8, 4, and 8, respectively. On the DOTA dataset, the batch sizes for RetinaNet, Faster RCNN, FCOS, YOLOv3, and YOLOX are 4, 1, 4, 4, and 4, respectively.

C. Comparison With SOTA Methods

We compare the proposed method with recent SOTA KD methods, i.e., FitNet [37], attention transfer (AT) [38], DeFeat [18], FRS [19], FGD [20], MGD [21], and instance-aware distillation (InsDist) [47] built upon the five benchmark detection networks. We conducted experiments over the DOTA dataset to compare the distilled detectors with current object detection methods in the remote sensing field, including Oriented RCNN [31], R3Det [32], S^2A-Net [33], ReDet [34], GWD [53], and ROI transformer [54]. The metrics of mean average precision (mAP), AP_s, AP_m, AP_l, AR_s, AR_m, AR_l, frame per second (FPS), parameters (Params), and floating point operations (FLOPs) are selected to assess the performance of different KD methods and different object detection methods in remote sensing images. mAP is the mean of the average precision of all categories. AP_s and AP_m, AP_l and AR_s, and AR_m and AR_l denote the average precision and average recall for small, medium, and large targets, respectively. Params and FLOPs denote the number of parameters and computation of the model, and FPS denotes the number of image frames processed by the model per second.

1) Quantitative Comparison: In order to conduct a fair comparison, the hyperparameters for all comparative algorithms remain consistent with their respective original papers. Since some algorithms were not proposed over YOLOv3 and YOLOX, we conducted experiments using hyperparameters consistent with other detectors. W_{bg} and W_{fg} are set to $\{1, 1\}$ and $\{4, 0.6\}$ for Faster RCNN and YOLOv3. For the other three detectors, W_{bg} and W_{fg} are set to $\{2, 0.6\}$. α are set to 5×10^{-5} for RetinaNet, 7×10^{-5} for Faster RCNN, 1×10^{-5} for FCOS, 2×10^{-5} for YOLOv3, and 3×10^{-10} for YOLOX, respectively.

Table I presents the comparison results of the teacher model, the student model, and the distilled student model in terms of mAP, Params, FPS, and FLOPs. Compared with the teacher model, the student model has significantly lower Params and FLOPs as well as higher FPS for detection. After training with ACFG, the FLOPs of the student model remain the same, and the FPS remains essentially the same or has a one-frame improvement, but the performance is improved. For the DIOR dataset, the mAP of the distilled student models for RetinaNet, Faster RCNN, FCOS, YOLOv3, and YOLOX have been improved by 3.9%, 2.4%, 4.6%, 6.8%, and 1.1%, respectively. For the DOTA dataset, the mAP of the distilled student models for RetinaNet, Faster RCNN, FCOS, YOLOv3, and YOLOX have been improved by 2.7%, 1.1%, 19.7%, 1.1%, and 1.1%, respectively. It can be seen that the performance of the distilled student model is significantly improved, approaching or even surpassing that of the teacher model, while maintaining a lightweight model.

Tables II and III show the results of the teacher model, the student model, and the student model trained with different KD methods over DIOR and DOTA datasets, including

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TABLE II
COMPARISON WITH SOTA OBJECT DETECTION KD METHODS ON THE DIOR

Method	mAP	AP_s	AP_m	AP_l	AR_s	AR_m	AR_l
RetinaNet (T)	60.9	5.0	30.5	61.9	9.8	40.2	71.0
RetinaNet (S)	56.6	4.9	27.7	56.0	9.8	39.5	67.7
AT [38]	60.1	5.5	**31.6**	61.8	10.9	**41.7**	71.3
DeFeat [18]	58.6	4.5	29.1	60.0	9.6	39.3	69.2
FGD [20]	56.3	4.3	27.9	56.6	9.3	38.1	67.0
FitNet [37]	60.1	4.9	30.4	**62.2**	9.7	39.7	71.3
FRS [19]	60.0	5.5	**31.6**	61.8	10.9	**41.7**	71.3
InsDist [47]	58.5	4.3	28.5	60.3	9.0	38.0	69.4
MGD [21]	58.1	4.4	28.4	59.3	9.4	38.1	67.0
ACFG	**60.5**	**5.6**	31.3	61.5	**11.5**	41.4	**71.4**
Faster RCNN (T)	60.5	5.9	29.1	61.4	9.9	39.1	69.6
Faster RCNN (S)	58.0	5.4	27.5	55.6	7.4	33.5	62.8
AT [38]	59.9	5.9	30.1	60.2	8.9	39.0	69.2
DeFeat [18]	58.2	5.4	27.5	57.7	7.6	34.1	66.1
FGD [20]	58.0	5.5	27.6	55.1	8.4	35.0	64.6
FitNet [37]	60.3	4.9	**30.4**	**62.3**	8.7	39.2	**71.7**
FRS [19]	59.0	5.8	28.6	59.4	7.8	34.2	67.4
InsDist [47]	59.4	5.9	28.6	59.6	8.1	34.8	67.4
MGD [21]	58.2	5.5	28.1	57.3	7.4	34.0	65.2
ACFG	**60.4**	**6.3**	**30.4**	60.5	**9.4**	**39.4**	69.0
FCOS (T)	63.1	6.2	32.9	61.6	12.8	45.6	73.4
FCOS (S)	58.2	5.2	28.0	56.4	10.5	40.7	68.7
AT [38]	62.0	5.6	28.9	58.3	11.8	41.5	70.6
DeFeat [18]	55.2	5.6	26.6	53.3	10.5	44.0	67.6
FitNet [37]	61.4	6.6	**31.9**	60.7	11.8	**44.7**	72.4
FRS [19]	61.1	5.8	29.4	58.9	11.7	42.7	70.7
InsDist [47]	52.5	4.9	25.3	49.3	9.7	38.9	65.8
MGD [21]	59.6	5.7	29.7	58.7	11.3	42.0	71.5
ACFG	**62.8**	**6.7**	31.0	**61.8**	**12.3**	43.9	**72.9**
YOLOv3 (T)	61.6	7.8	28.8	55.9	12.3	36.6	64.7
YOLOv3 (S)	49.3	6.1	21.7	37.7	11.5	29.2	49.3
AT [38]	55.8	7.3	24.7	43.9	12.6	32.9	55.3
DeFeat [18]	55.8	7.1	24.7	**44.6**	12.9	33.1	**55.7**
FGD [20]	54.6	6.9	24.3	43.8	**13.6**	33.0	55.3
FitNet [37]	56.0	6.9	24.6	44.3	13.1	32.8	55.2
FRS [19]	55.3	7.1	24.5	43.9	12.9	33.1	54.9
InsDist [47]	55.5	7.2	**24.8**	44.0	12.9	33.1	55.1
MGD [21]	55.6	7.1	**24.8**	44.0	12.7	**33.5**	54.6
ACFG	**56.1**	**7.4**	24.7	**44.6**	13.0	**33.5**	55.4
YOLOX (T)	73.9	15.3	40.5	69.9	21.3	48.0	76.2
YOLOX (S)	70.8	14.3	38.5	63.0	21.0	47.2	70.8
AT [38]	71.2	13.9	38.6	**64.9**	19.9	46.6	**72.1**
DeFeat [18]	71.5	14.0	**38.9**	**64.9**	20.5	**46.7**	**72.1**
FGD [20]	67.9	12.4	35.9	62.1	18.3	44.8	70.5
FitNet [37]	71.0	14.1	38.4	64.8	20.2	**46.7**	**72.1**
FRS [19]	71.0	14.0	38.7	64.8	20.7	47.3	71.9
InsDist [47]	69.2	13.1	36.5	62.8	18.5	44.6	70.2
MGD [21]	69.8	13.4	37.2	63.2	19.6	46.1	70.9
ACFG	**71.9**	**14.6**	38.7	64.8	**21.3**	**46.7**	71.8

TABLE III
COMPARISON WITH SOTA OBJECT DETECTION KD METHODS ON THE DOTA

Method	mAP	AP_s	AP_m	AP_l	AR_s	AR_m	AR_l
RetinaNet (T)	50.3	6.5	33.1	45.6	13.2	43.0	54.0
RetinaNet (S)	47.9	6.3	31.5	43.1	12.1	40.7	52.8
AT [38]	47.0	4.8	29.9	42.1	10.0	40.1	54.1
DeFeat [18]	48.2	5.1	31.3	42.6	11.4	40.4	51.7
FGD [20]	46.6	4.6	29.5	39.8	10.0	39.2	50.6
FitNet [37]	49.8	5.2	32.3	44.7	12.5	41.8	53.6
FRS [19]	50.3	**5.6**	33.0	44.4	12.5	42.4	53.4
InsDist [47]	48.6	5.4	31.1	44.4	11.3	40.3	53.4
MGD [21]	47.4	4.8	30.3	42.9	11.7	41.7	53.7
ACFG	**50.6**	**5.6**	**33.1**	**45.1**	**12.9**	**42.9**	**54.9**
Faster RCNN (T)	52.6	10.2	36.7	45.7	13.6	44.3	52.7
Faster RCNN (S)	52.0	10.1	36.4	44.6	14.5	44.0	52.4
AT [38]	52.6	11.5	**37.4**	**46.1**	14.9	45.2	53.4
DeFeat [18]	52.7	11.3	**37.4**	45.0	15.1	**46.2**	**54.4**
FGD [20]	49.6	8.6	33.5	39.5	11.8	42.3	50.0
FitNet [37]	52.7	**11.9**	37.0	44.2	15.2	45.4	52.7
FRS [19]	52.8	11.7	36.9	43.7	15.4	45.5	52.5
InsDist [47]	52.7	11.5	37.3	44.3	14.6	46.0	53.3
MGD [21]	52.0	9.5	34.6	40.9	12.8	43.3	49.8
ACFG	**53.1**	**11.9**	**37.4**	44.2	**15.5**	**46.2**	53.6
FCOS (T)	53.6	10.5	33.5	42.9	17.2	43.8	53.2
FCOS (S)	33.5	5.3	23.2	20.6	9.3	36.3	40.1
AT [38]	52.4	10.0	34.0	44.1	**18.3**	45.4	56.0
DeFeat [18]	48.2	8.8	31.3	38.8	15.0	43.6	51.5
FitNet [37]	52.5	9.8	33.7	43.6	17.5	46.0	54.7
FRS [19]	52.7	9.9	**34.3**	43.9	17.5	46.2	55.5
InsDist [47]	45.2	8.3	29.6	35.2	15.2	41.6	50.2
MGD [21]	49.9	9.6	32.9	41.3	17.1	45.9	53.7
ACFG	**53.2**	**10.2**	34.0	**45.2**	17.5	**46.3**	**58.6**
YOLOv3 (T)	56.4	13.0	35.1	41.3	19.6	43.9	49.2
YOLOv3 (S)	50.9	11.9	30.7	31.6	20.5	41.9	41.2
AT [38]	51.0	11.1	30.5	31.7	18.2	41.8	40.8
DeFeat [18]	51.2	11.3	30.3	31.8	18.9	41.6	41.8
FGD [20]	50.5	10.9	30.1	31.8	18.4	41.8	**41.9**
FitNet [37]	50.9	11.6	30.5	**32.8**	19.8	41.9	**41.9**
FRS [19]	51.1	11.6	30.6	32.4	**20.1**	**42.0**	41.2
InsDist [47]	51.1	**11.9**	30.4	32.6	19.2	41.8	41.7
MGD [21]	51.4	11.4	30.1	31.6	19.2	41.8	41.6
ACFG	**52.0**	**11.9**	**30.7**	32.2	19.2	**42.0**	41.3
YOLOX (T)	66.2	23.8	44.6	50.2	31.4	55.0	58.6
YOLOX (S)	64.0	21.9	41.9	44.5	29.3	52.5	55.3
AT [38]	64.0	21.5	42.3	48.9	27.7	51.6	57.9
DeFeat [18]	64.2	21.1	42.6	48.7	28.8	**53.3**	**58.2**
FGD [20]	64.3	21.3	41.8	48.5	27.7	52.1	57.9
FitNet [37]	64.1	21.1	42.3	48.9	28.0	51.7	58.0
FRS [19]	63.9	21.0	42.1	48.7	28.1	52.8	57.8
InsDist [47]	64.4	21.4	42.7	**49.0**	28.7	52.8	57.9
MGD [21]	57.8	17.0	35.1	38.4	24.2	47.4	52.1
ACFG	**65.1**	**21.7**	**42.8**	48.5	**28.9**	52.3	57.7

mAP, AP_s, AP_m, AP_l, AR_s, AR_m, and AR_l. For the DIOR dataset, compared with the other KD methods, the proposed method improves the mAP of RetinaNet, Faster RCNN, FCOS, YOLOv3, and YOLOX by 0.4%, 0.1%, 0.8%, 0.1%, and 0.4%, respectively. For the DOTA dataset, compared with the other KD methods, the proposed method improves the mAP of RetinaNet, Faster RCNN, FCOS, YOLOv3, and YOLOX by 0.3%, 0.3%, 0.5%, 0.6%, and 0.7%, respectively. It can be found that our ACFG obtains the highest values of mAP, AP_s, and AR_s, which effectively improves the target detection results, especially for small targets.

The reasons why our method can achieve better performance are analyzed as follows. Some of the previous algorithms, e.g., MGD, FitNet, and AT, apply convolution or normalization over the entire feature map and then compute the loss, however, they ignore the differences between foreground and background information and do not take full advantage of global correlation. Moreover, many other algorithms typically employ multiple branches for distillation, such as FGD, DeFeat, and InsDist, which are greatly affected by the weighting hyperparameters. Therefore, we embed foreground and background information as well as global and local information in CMG and GLPL to achieve adaptive feature transfer in a single-branch structure. First, CMG uses the foreground masks to refine rich target information and the random masks to provide background information, which acts as a kind of contextual information to provide additional hints for detecting the target. CMG makes full use of foreground information for localization and background information for assisted recognition. Second, the GLPL combines the self-attention mechanism and DCN

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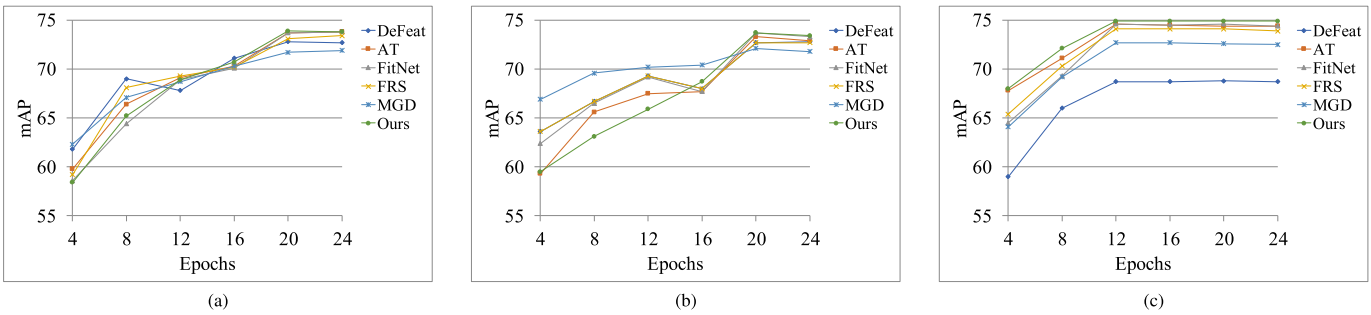


Fig. 4. mAP during training with different distillation methods on the DIOR. (a) RetinaNet. (b) Faster RCNN. (c) FCOS.

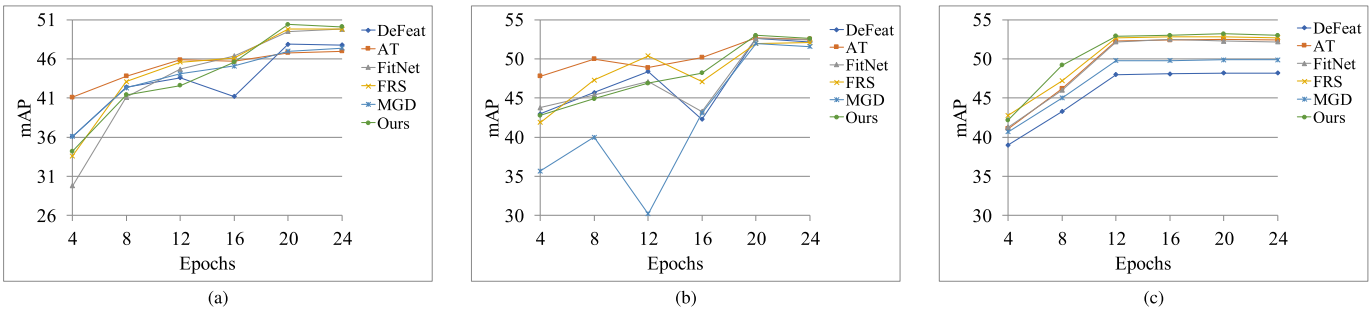


Fig. 5. mAP during training with different distillation methods on the DOTA. (a) RetinaNet. (b) Faster RCNN. (c) FCOS.

for feature mapping, which allows the network to better learn the global correlation as well as the target shape features. Moreover, GLPL acts as an intermediate layer capable of mitigating the overfitting between the two feature maps. As a result, CMG and GLPL adaptively learn global and local features and enhance the robustness of student feature points. In addition, we incorporate target sizes into loss computation to enhance the focus on small targets, thus improving the performance of the student model.

Table IV presents the results of the distilled algorithm with other target detection algorithms in remote sensing images in terms of mAP, FLOPs, FPS, and Params. We set the number of training epochs to 24 and the image size to 640 × 640 for fair comparison. In general, the performance of the five student detectors after ACFG mostly exceeds the performance of the target detectors designed for remote sensing images, and the detection speed is comparable or faster. Specifically, the distilled RetinaNet, Faster RCNN, and FCOS have similar Params and FLOPs to the compared remote sensing target detection models, but the performance is improved, especially the mAP of the distilled FCOS is improved by at least 0.5%. The distilled YOLOv3 and YOLOX have fewer parameters and higher detection speeds, basically double the detection speed of the compared algorithms. Although the performance of YOLOv3 decreases, when the epoch is increased by a small amount, it achieves an mAP of 52.0% with 3.68M and 6.61 GFLOPs. YOLOX has comparable performance after 24 epochs of distillation training, which achieves an mAP of 42.6% with 8.94M and 13.34 GFLOPs. With our ACFG, detectors with different accuracies and inference speeds are presented, enhancing the utility of lightweight networks.

Previous KD algorithms have shown different results on different object detectors and have not achieved the best

TABLE IV
COMPARISON WITH SOTA OBJECT DETECTION METHODS ON THE DOTA

Method	epoch	mAP	FPS	Params (M)	FLOPs (G)
Oriented RCNN [31]	24	52.0	17	41.14	91.07
R3Det [32]	24	49.5	13	41.30	131.34
S^2A-Net [33]	24	52.7	14	38.60	77.23
ReDet [34]	24	42.0	10	31.65	48.35
GWD [53]	24	49.4	16	36.44	84.52
ROI transformer [54]	24	52.3	16	55.13	105.13
ACFG+RetinaNet	24	50.6	19	36.42	84.34
ACFG+Faster RCNN	24	53.1	19	41.20	91.08
ACFG+FCOS	24	**53.2**	16	32.87	59.22
ACFG+YOLOv3	24	34.1	**36**	**3.68**	**6.61**
ACFG+YOLOX	24	42.6	35	8.94	13.34
ACFG+YOLOv3	30	52.0	36	3.68	6.61
ACFG+YOLOX	300	65.1	35	8.94	13.34

performance on multiple object detectors. Figs. 4 and 5 show the mAP curves during the training process of different algorithms on the DIOR and DOTA datasets, respectively. The training trends of these algorithms are roughly the same. Compared with other algorithms, our ACFG shows an overall upward trend as the training period increases, and the downward trend and turning point are smaller, showing that the overall algorithm is more stable. Around the 20th epoch, our ACFG achieves the best performance and stabilizes at the 24th epoch. Overall, our ACFG has outperformed commonly used KD methods on both the DIOR and DOTA datasets, achieving SOTA performance. In some cases, certain student models have even outperformed the teacher model. This demonstrates the effectiveness of this algorithm in processing complex remote sensing images. Due to the utilization of the projection layer for implementing adaptive distillation, the student model is compeled to generate teacher features with masked characteristics. This process enhances the representation capability

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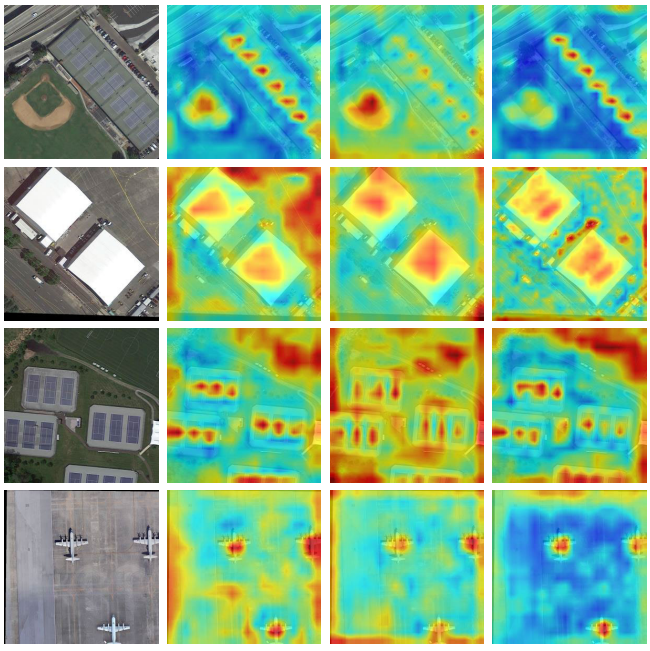


Fig. 6. Visualization of the feature from the first layer of FPN outputs.

of the student model. As a result, our ACFG has achieved a notable increase in mAP.

2) Qualitative Analysis: In order to validate that our proposed algorithm enhances the feature representation capability of the student model, we conducted visualizations of the neck features for both the student and teacher models. Fig. 6 shows the visualization of attention maps derived from different models. We conduct feature map visualization and analysis based on YOLOv3, MobileNetV2, and DarkNet53 are employed as the student and teacher model backbones, respectively. Featuremap visualization provides insights into the learning process and image features that different models focus on. Through feature map visualization, we can better understand what the model focuses on and how the student model performs. As is shown in Fig. 6, the student and teacher detectors exhibit significant differences in the features they attend to. Compared with the teacher detector, the student detector demonstrates a higher response to background information. After training with ACFG, there still exist differences in the features, but the student's response to the background has significantly decreased. In addition, the student model with ACFG locates the center of the object more accurately and enhances the response to small targets, which helps to improve the detection performance.

Figs. 7 and 8 show the detection results of the student using RetinaNet trained with ACFG on the DIOR and DOTA, respectively. The distilled student model shows good detection performance on dense and multiscale objects. Fig. 9 shows the results achieved by the plain FCOS detector and the distilled FCOS detector. We use FCOS with ResNet50 as the base model and distill the base model using our ACFG with a ResNet101-based large model as the teacher. It is observed that: 1) the base model misses instances in naturally noisy images, while the distilled detector can accurately detect ships, as is shown in Fig. 9(a). 2) Compared with the base model, the distilled model removes redundant detection boxes and detects missed objects, as is shown in Fig. 9(b) and (c). 3) With our ACFG for distillation, detection performance for small instances, such as the small vehicles in Fig. 9(d) and the storage tanks in Fig. 9(e) are enhanced.

TABLE V
EFFECT OF EACH MODULE IN ACFG ON DIOR DATASET

Detector	CMG	GLPL	BDL	mAP
RetinaNet				59.7
	✓			60.2
	✓	✓		60.5
	✓	✓	✓	**60.5**
FCOS				60.8
	✓			62.3
	✓	✓		62.8
	✓	✓	✓	**62.8**
YOLOv3				51.8
	✓			55.8
	✓	✓		56.1
	✓	✓	✓	**56.1**

D. Discussion

In this section, we first conduct ablation experiments on the CMG module, GLPL module, and different distillation losses to compare their effects. Second, we discuss the impact of each hyperparameter. The purpose of these experiments is to validate the contributions of the individual modules of our proposed algorithm.

1) Effect of Each Module: We evaluated the effectiveness of each module on the DIOR datasets, including the CMG module, GLPL module, and BDL. Table V has shown the results of an ablation study based on RetinaNet, FCOS, and YOLOv3. Each module of our proposed algorithm has a significant effect on the performance of the student model, and we conduct more detailed experiments and in-depth analysis later.

2) CMG Module: We perform experiments to assess the impact of the random mask branch, instance mask branch, and background mask branch in the CMG module by comparing their inclusion and exclusion. Our CMG module uses multiple masks to select student features and forces it to generate a complete feature map of the teacher which is different from most methods. This enables us to evaluate the contributions of these components within the framework. We adjust the branch hyperparameter β and set it to 0.5.

We conduct experiments on RetinaNet and FCOS detectors using different feature projection models, where the CNN contains two layers of conv and one layer of relu, and GLPL is our proposed module. BG and FG denote background and instance branches, respectively. Table VI presents the experimental results on the DIOR dataset. The results show that both random masks and instance or background masks play important roles, which can be effectively combined to improve learning performance. The student detector effectively directs the attention of the student detector to important components through random masks and instance masks. The additional mask branches all improve the efficiency of our model, especially the instance masks. The target area on

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Fig. 7. Sample detection results of the student using RetinaNet detector trained with ACFG on the DIOR.



Fig. 8. Sample detection results of the student using RetinaNet detector trained with ACFG on the DOTA.

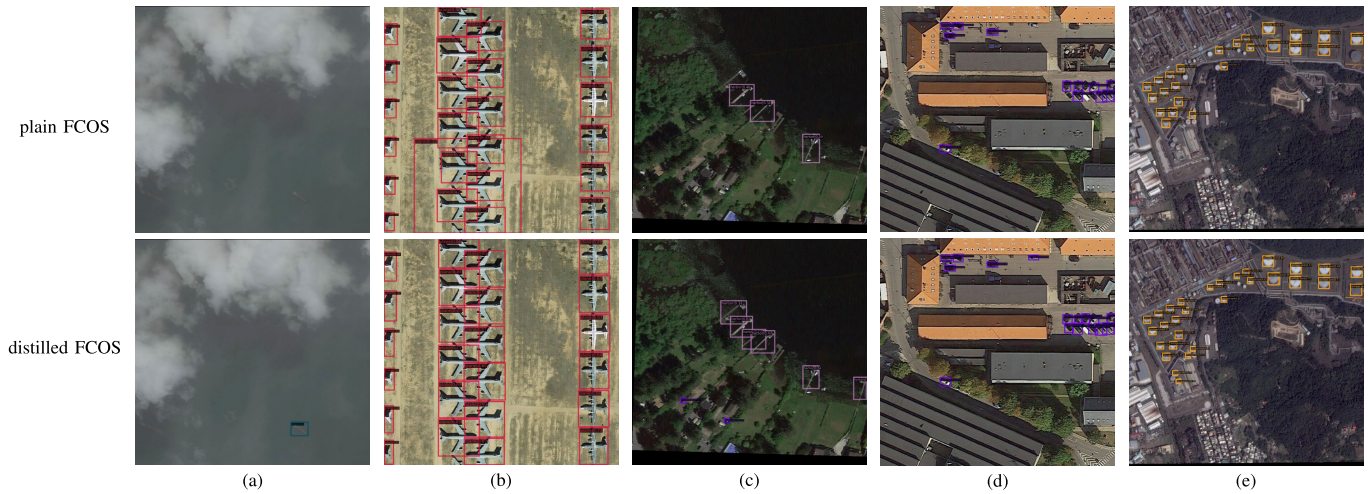


Fig. 9. Qualitative comparison between the plain detector and detector trained with our ACFG on the DOTA. (a) Natural noise. (b) Redundant detection box. (c) Missed harbor targets. (d) Small vehicles. (e) Small storage tanks.

the feature map can provide more instance information for the detection task, allowing the student model to focus on learning the characteristics of the instances, thus affecting the distillation effect.

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TABLE VI
EFFECT OF DIFFERENT BRANCHES IN CMG ON DIOR DATASET

Detector	Random	BG	FG	mAP
RetinaNet+CNN	✓			58.1
	✓	✓		58.7
	✓		✓	**59.7**
RetinaNet+GLPL	✓			58.7
	✓	✓		59.1
	✓		✓	**60.2**
FCOS+CNN	✓			59.6
	✓	✓		60.5
	✓		✓	**60.8**
FCOS+GLPL	✓			60.3
	✓	✓		61.9
	✓		✓	**62.3**

TABLE VII
EFFECT OF DIFFERENT PROJECTION LAYER

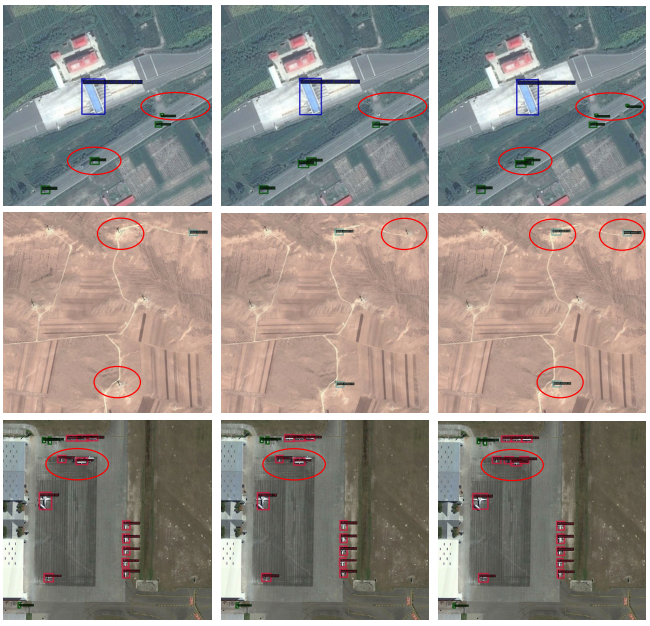
Detector	CNN	RES	GLPL
RetinaNet	59.7	59.6	**60.2**
YOLOv3	51.8	51.8	**55.8**
FCOS	60.8	60.1	**62.3**

TABLE VIII
EFFECT OF DIFFERENT LOSS

Detector	MSE			DFL			BDL		
	mAP	AP_s	AR_s	mAP	AP_s	AR_s	mAP	AP_s	AR_s
RetinaNet	60.2	5.1	10.8	60.4	**5.9**	11.0	**60.5**	5.6	**11.5**
Faster RCNN	59.2	5.3	7.6	59.6	**6.3**	9.2	**60.4**	**6.3**	**9.4**
YOLOv3	55.8	7.2	12.8	55.8	7.0	**13.0**	**56.1**	**7.4**	**13.0**
FCOS	62.3	6.5	12.2	62.6	6.6	**12.5**	**62.8**	**6.7**	12.3

3) GLPL Module: The masked feature distillation generation layer is generally a conv layer, but it can only extract local features and does not have global recovery properties. We conducted comparative experiments to verify the excellent mask generation characteristics of the GLPL layer. The projection layer is usually a simple structure used to realize the mapping between two feature maps. An overly complex projection layer contains a large number of parameters, which weakens the connection between the two feature maps. We conduct comparative experiments on three different projection layers. In addition to CNN, we also experiment with the residual structure that contains two layers of conv, one layer of relu, and a short connection. Table VII has shown the result of different projection layers. GLPL has the best performance due to its good global feature extraction capabilities, improving the performance of lightweight student models.

4) Different Distillation Loss: Our proposed BDL is compared with two commonly used feature distillation loss functions, including mean squared error (MSE) and decoupled feature loss (DFL). Four detectors are adopted, including RetinaNet, Faster RCNN, YOLOv3, and FCOS. As shown in Table VIII, our BDL outperforms MSE and DFL for all the target detectors since it comprehensively takes into account the differences between foreground and background, as well as the size characteristics of the instances. It can be seen that the detection precision and recall of small targets, including



MSE DFL BDL

Fig. 10. Sample detection results of different loss using YOLOv3 detector on the DIOR.

AP_s and AR_s, are improved, which verifies the effectiveness of BDL for small targets. Fig. 10 presents the detection results of the YOLOv3 detector with different loss functions, and BDL detects missed small targets such as vehicles, windmills, and airplanes, which are emphasized with red ellipses.

5) Sensitivity to Hyperparameters: We conduct a comparative experimental study on the rationality of the hyperparameters. The proposed algorithm utilizes five key hyperparameters. W_{fg} and W_{bg} are used to balance the distillation loss related to the foreground and background, and α is used to adjust the proportion of distillation loss in the overall loss. λ represents a hyperparameter that sets the threshold for the random mask branch. β denotes the hyperparameter adjusting the random and foreground branch. In order to better validate the rationality of the hyperparameters, we have presented the curves of α for RetinaNet, Faster RCNN, and YOLOv3 in Fig. 11, and the curves of β, W_{bg} and λ for RetinaNet in Fig. 12. As α increases, the mAP increases gradually and reaches a maximum value for the reason that the student model can learn the features of the teacher model, which improves the performance of the student. However, if α continues increasing, the mAP decreases significantly because the student model begins to focus excessively on the distillation loss, which diminishes the influence of the original student loss, leading to a decrease in distillation effectiveness. Fig. 12(a), demonstrates that the best result is achieved when both branches obtain the same weight of 0.5. As β increases, the student model could utilize the background information for assisted recognition leading to an increase in mAP. However, if β continues increasing, the student model focuses on the background information and ignores the target information, resulting in a decrease in mAP. From Fig. 12(b), it can be found that the trend of W_{bg} is almost the same with α. If W_{bg} is too small, the background boundary information will be blurred, while if W_{bg}

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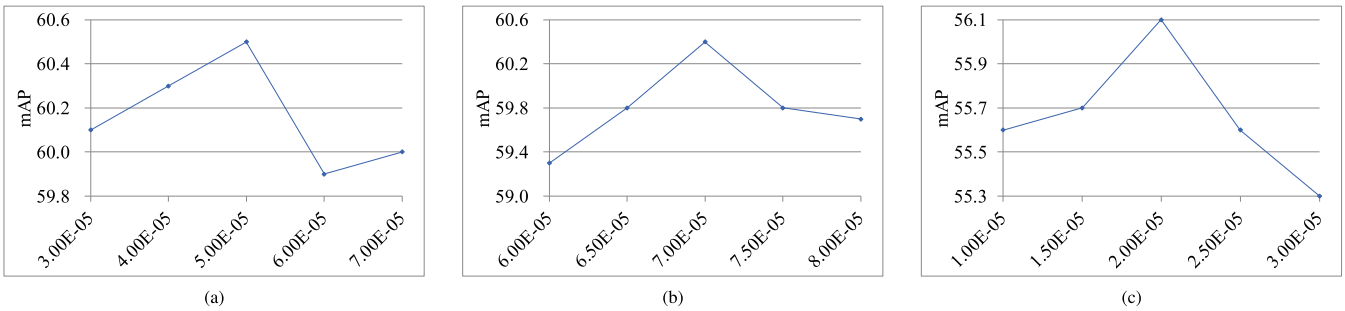


Fig. 11. Hyperparameter sensitivity study of α on the DIOR dataset. (a) RetinaNet. (b) Faster RCNN. (c) YOLOv3.

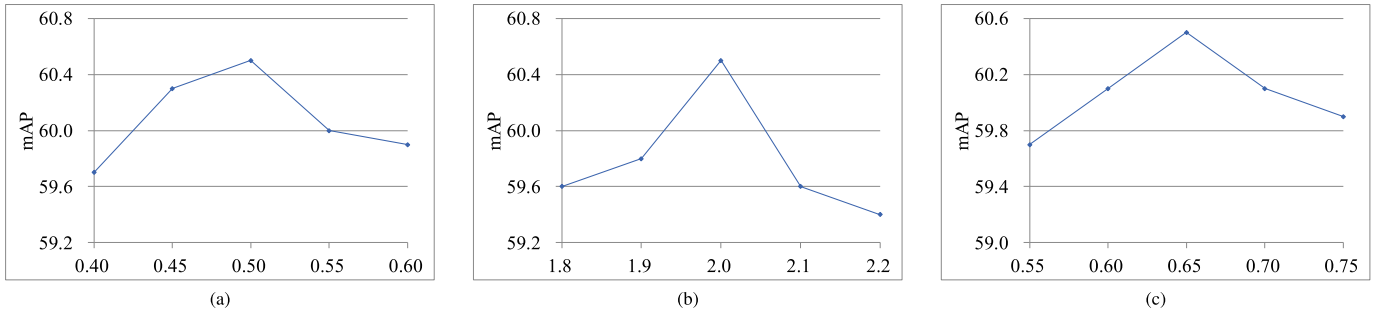


Fig. 12. Hyperparameter sensitivity study of β, W_{bg}, and λ on the DIOR dataset. The teacher and student models are RetinaNet based on ResNet101 and ResNet50, respectively. (a) Influence of β. (b) Influence of W_{bg}. (c) Influence of λ.

is too large, it will pay more attention to a large amount of background information and miss information related to the instances. Fig. 12(c) also presents a trend of first increasing and then decreasing, we can find that when λ is too small, the random mask selects fewer feature points, thus making features difficult to recover, when λ is too large, more feature points introduce additional interference information, making the detection effect worse.

E. Impact of Different Student and Teacher Detectors

In this section, the student model is fixed and different teacher models are used for distillation to evaluate the impact of performance differences between the two on distillation results.

The existing research has found that a high-performing teacher model in image classification tasks does not necessarily benefit the distillation of the student model [55]. When there is a significant performance gap between the teacher and student models, it can lead to a decrease in performance. First, we employed RetinaNet based on ResNet18 as the student model and conducted KD using various teacher models. As shown in Table IX, there is no positive correlation between the performance of the teacher model and the student model, and the distillation effect decreases with the gap between the models widening. Second, we conduct experiments on heterogeneous backbones. It is well known that distillation performance is hindered when using heterogeneous backbones [56], [57]. As shown in Table X, we select MobileNetV2 as the backbone of the student, and MobileNetV2, ResNet50, and DarkNet53 as teachers, and the performance has been improved by 1.6%, 0.2%, and 1.1%, respectively. These findings on KD in object detection are consistent with the observations made in image classification.

TABLE IX
DISTILLING A STUDENT WITH DIFFERENT TEACHERS ON DOTA. RESNET18 IS THE BACKBONE FOR THE STUDENT

Teacher	ResNet18	ResNet50	ResNet101
	40.3	47.9	50.3
*Student		40.3	
+ACFG	47.1 (+6.8)	**47.2 (+6.9)**	45.4 (+4.1)

TABLE X
DISTILLING A STUDENT WITH DIFFERENT TEACHERS ON DOTA. MOBILENETV2 IS THE BACKBONE FOR THE STUDENT

Teacher	MobileNetV2	ResNet50	DarkNet53
	50.9	51.9	56.4
*Student		50.9	
+ACFG	**52.5 (+1.6)**	51.1 (+0.2)	52.0 (+1.1)

Our design of remote sensing object detection operates on intermediate high-level features. As the network model deepens, the information difference between the features produced by the two models increases and the learning of informative features becomes difficult. In addition, our proposed algorithm has good performance in self-distillation. We select ResNet18 and MobileNetV2 as teacher models and their mAP have been improved by 6.8% and 1.6%, respectively, which can be seen from Tables IX and X.

F. Other Discussions

1) Ground Truth: There are indeed limitations in generating masks based on ground truth. We will continue to investigate how to generate masks without ground truth, such as using simple threshold segmentation or introducing some segmen-

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tation algorithms to generate different masks based on global correlation.

2) Image Modality: Our ACFG has no explicit restrictions on the inputs and deals with the features between the layers of the network. Theoretically, ACFG is also applicable to target detection networks for SAR images, hyperspectral images, and multispectral images, which can improve the performance of student models. However, the level of performance of models trained with ACFG in their fields is uncertain, which is largely limited by the choice of teacher and student models. In the future, we will validate the proposed method on target detection networks for different data to further improve the method's adaptability to different modalities.

3) Distillation Loss: In order to achieve more accurate knowledge transfer between teacher and student models, we will work on introducing other information such as instance category information and confidence value in the computation of the loss function.

V. Conclusion

In this article, we proposed an ACFG method for efficient object detection in remote sensing images. ACFG contains three important modules: a CMG module, a GLPL module, and BDL as KD loss. The CMG module builds a parallel branch, randomly selects student features, and adds instance information. With the ability to restore the complete image with few feature points during the training process, it improves the robustness of each feature point on the student feature map, so that the student model has better feature generation capabilities. GLPL adaptively realizes the global and local mapping of the feature maps using the attention mechanism and DCN. BDL introduces the size of instances and decouples the foreground and background, respectively, to calculate the distillation loss. Experiments on two large-scale datasets of DIOR and DOTA show that our algorithm's two-stage and one-stage, as well as anchor-based and anchor-free object detectors, show good versatility. In the future, we will continue to investigate the generation of masks without ground truth, and new loss functions to achieve effective transfer, as well as validate the applicability of the proposed algorithms to different modal data.

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